Digital Electronic Circuits (EC2L004)

Lecture 1: Binary Numbers – Part I

Learning Objectives:

- Binary numbers
- Number-base conversions
- Octal and Hexadecimal Numbers
- Complement of Numbers

Decimal Numbers: Recap

- Digits : {0,1,2,3,4,5,6,7,8,9}
  - How to represent numbers greater than 9 ?
  - Place-value system
  - Example:

$$7{,}392.67 = 7\text{x}10^{3} + 3\text{x}10^{2} + 9\text{x}10^{1} + 2\text{x}10^{0} + 6\text{x}10^{-1} + 7\text{x}10^{-2}$$

  - Base 10 system or Radix 10
- Can we define arbitrary large numbers using fewer or more digits?
  - Examples:
    - Binary-digits = { 0,1}
    - Octal-digits = {0,1,2,3,4,5,6,7}
    - Hexadecimal-digits ={0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}

Base-r system

Let the digits of a base-r system be $\mathcal{B} = \{0, 1, 2, \ldots, r-1\}$.
A base-r number

$$(a_n a_{n-1} \cdots a_0 . a_{-1} a_{-2} \cdots a_{-m})_r$$

where $a_i \in \mathcal{B}$ is equal to decimal number:

$$a_n * r^n + a_{n-1} * r^{n-1} + \cdots + a_1 * r + a_0 + a_{-1} r^{-1} + a_{-2} * r^{-2} + \cdots + a_{-m} * r^{-m}$$

- Convert the following numbers to decimal?
  - $(4021.2)_5 = (511.4)_{10}$
  - $(127.4)_8 = (87.5)_{10}$
  - $(B65F)_{16} = (46{,}687)_{10}$
  - $(110101)_2 = (53)_{10}$

  Note: Digits a binary number are called *"bits"*

Number-Base Conversions

- Representations of a number in a different radix are said to be equivalent if they have the same decimal representation.

- Example: $(0011)_8 = (1001)_2$

- The conversion of a number in base *r* to decimal is done by expanding the number in a power series and adding all the terms.

- How to convert a decimal number to a number in base *r* ?

Examples:

- Convert decimal 41 to binary.
- Convert decimal 153 to octal.
- Convert $(0.6875)_{10}$ to binary.
- Convert $(0.513)_{10}$ to octal.
- $(41.6875)_{10} = (101001.1011)_2$
- $(153.513)_{10} = (231.406517)_8$

Octal and Hexadecimal Numbers

- Numbers with different Bases

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Binary to Octal Conversion

- Partition the binary number into groups of three digits each, starting from the binary point.

- Proceed to the left and to the right and assign the corresponding octal digit to each group.

- Example:

$$\begin{array}{cccccccccc} (10 & 110 & 001 & 101 & 011 & \cdot & 111 & 100 & 000 & 110)_2 = (26153.7406)_8 \\ 2 & 6 & 1 & 5 & 3 & & 7 & 4 & 0 & 6 \end{array}$$

Binary to Hexadecimal Conversion

- Conversion from binary to hexadecimal is similar, except that the binary number is divided into groups of *four* digits:

- Example:

$$(10\ \ 1100\ \ 0110\ \ 1011\ \ \cdot\ \ 1111\ \ 0010)_2 = (2C6B.F2)_{16}$$

$$2\ \ \ \ C\ \ \ \ 6\ \ \ \ B\ \ \ \ F\ \ \ \ 2$$

Octal/Hexadecimal to Binary Conversion

- Each octal digit is converted to its three-digit binary equivalent

$$(673.124)_8 = (\underset{6}{110}\ \underset{7}{111}\ \underset{3}{011}\ \cdot\ \underset{1}{001}\ \underset{2}{010}\ \underset{4}{100})_2$$

- Each hexadecimal digit is converted to its four-digit binary equivalent.

$$(306.\mathrm{D})_{16} = (\underset{3}{0011}\ \underset{0}{0000}\ \underset{6}{0110}\ \cdot\ \underset{\mathrm{D}}{1101})_2$$

Complement of Numbers

- Complements are used in digital computers to **simplify the subtraction operation** and for logical manipulation.

- Simplifying operations leads to simpler, less expensive circuits to implement the operations.

- There are two types of complements for each base-*r* system:
  - Radix complement (*r*'s complement)
  - Diminished radix complement ((*r* - 1)'s complement)

Diminished Radix (r-1)'s Complements

- Given a number N in base-r having n digits, the (r-1)'s complement of N : $(r^n - 1) - N$

- For decimal numbers, $r = 10$ and $r - 1 = 9$, so the 9's complement of N is $(10^n - 1) - N$.

- Decimal Example: n = 6 (6 digit) => $10^6 - 1 = 999999$

The 9's complement of

546700 is 999999 - 546700 = 453299

012398 is 999999 - 012398 = 987601

Diminished Radix (r-1)'s Complements

- For Binary Number (r-1)'s Complement: r=2 => r-1=1

The 1's complement of N is $(2^n -1)-N$, if n = 4

=> $2^4=(10000)_2$ then $(2^n -1)-N = (1111)_2 - N$

Example: Binary (r-1)'s Example: n = 7 (7 digit)

=> $2^7 -1=(1111111)_2$

- The 1's complement of 1011000 is 0100111.
- The 1's complement of 0101101 is 1010010.

Note: 1's complement of a binary number is formed by changing 1's to 0's and 0's to 1's.

Radix Complements

- The r's complement of an n-digit number N in base r is defined as $r^n - N$ for $N \neq 0$ and as 0 for $N = 0$.

- Comparison of r's and (r-1)'s complement

r's complement:

$$r^n - N$$
$$= [(r^n - 1) - N] + 1$$
$$= (r-1)\text{'s complement} + 1$$

(r-1)'s complement:

$$(r^n - 1) - N$$

Radix Complements Cont'd…

- Examples of *r*'s complement

- 10's (*r*'s) complement of decimal

2389 => 7610 (9's complement) +1= 7611

012398 => 987601 +1= 987602

246700 => 753299 +1=753300

- 2's (*r*'s) complement of binary

1101100 => 0010011 +1= 0010100

0110111 => 1001000 +1= 1001001

Thank You !!!